AI for Product Development

Activity 07: Engineer a Prompt and Evaluation Set

C840 • v1.0

Create a structured prompt contract and a frozen evaluation set for a product-feedback agent.

1. Purpose and deliverable

Create a structured prompt contract and a frozen evaluation set for a product-feedback agent.

Required output: Versioned prompt, 20-case evaluation CSV and scoring rubric

Acceptance: The evaluation set covers all segments; scoring rules are reproducible; baseline failures are retained.

2. Folder inventory

brief.yaml — machine-readable scope and decision contract

prompt.md — bounded agent instruction

data/mock-data.csv and mock-context.json — synthetic evidence

solution/expected-output.json — trainer reference shape

scripts/verify_activity.py — deterministic completeness check

evidence/checklist.md — learner sign-off record

3. Detailed procedure

1. Read brief.yaml and restate the product decision, output and acceptance rule.

2. Inspect the CSV and JSON inputs; list source IDs, dates, contradictions and missing evidence.

3. Open prompt.md and select an approved agentic AI builder. Do not add live data or integrations.

4. Ask the agent to plan the workflow before producing the artifact. Check roles, tools, data and stop conditions.

5. Execute the activity-specific workflow: Define contract → Sample cases → Add edges → Freeze rubric → Run baseline → Record gaps.

6. Require citations to source IDs and label every inference or assumption.

7. Review the output at the human gate; compare it with the acceptance rule and reject unsupported claims.

8. Save the artifact and evidence inside a learner working copy of this folder.

9. Run python3 scripts/verify_activity.py and retain the JSON result.

10. Complete evidence/checklist.md and record PASS, REPAIR or STOP with a named owner.

4. Troubleshooting

Missing citations: require row-level source IDs and rerun only the unsupported section.

Unstable recommendation: freeze criteria, rerun the same input and perform sensitivity analysis.

Verifier failure: compare the missing file or field with brief.yaml, repair it and rerun.

Unsafe action proposed: stop, remove live/external access and return the decision to the human owner.

5. Verification

Pass condition: The evaluation set covers all segments; scoring rules are reproducible; baseline failures are retained.

The deterministic verifier checks package completeness; the human reviewer checks evidence quality and decision fitness.